



Inspiring Excellence

CSE370 : Database Systems
Project Report
Project Title : MedKotha

Group No : 6 , CSE370 Lab Section : 17, Spring 2026		
ID	Name	Contribution
23301302	Baktier Galib	FT 1-3
23301489	Tahsin Pathan	FT 4-6
23301334	Shakeel	FT 7-9

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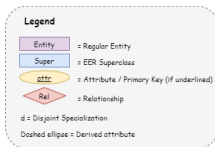
Introduction

MedKotha is a web based medical study platform aimed at helping Bangladeshi students learn their curriculum through active recall using practice questions, and flashcards rather than passive reading alone. The core idea is to combine a large question bank with explanations, an adaptive flashcard system, performance analytics on a personal dashboard, and optional social motivation through a leaderboard, so that learners can see progress, fix repeated mistakes, and review content in a structured way. The product also includes administration features so admins can maintain shared questions and flashcards and manage users. This keeps the learning material consistent for the whole process.

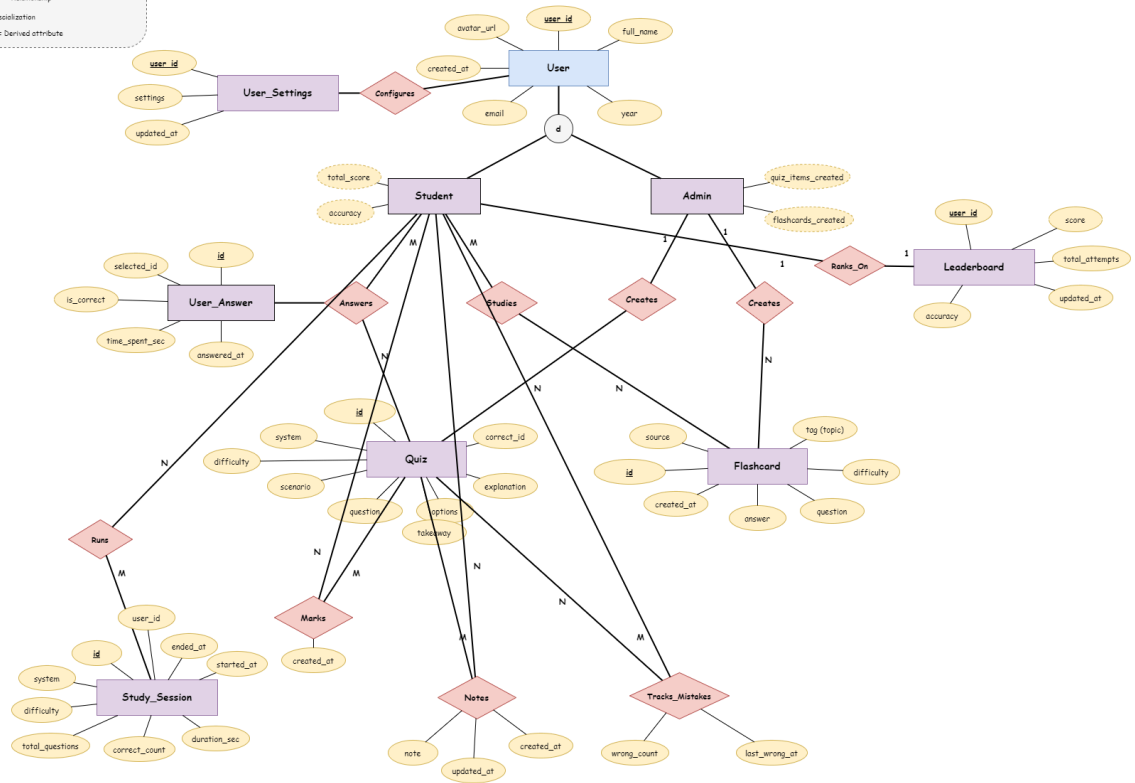
Project Features

ID, Name	Features [3 per member]	
23301302, Baktier Galib	Ft 1	Flashcards Management
	Ft 2	Quiz Management
	Ft 3	Users Management
23301489, Tahsin Pathan	Ft 4	Mistakes
	Ft 5	Leaderboard
	Ft 6	Admin Overview
23301334, Shakeel	Ft 7	Student Overview
	Ft 8	Flashcards
	Ft 9	Quiz

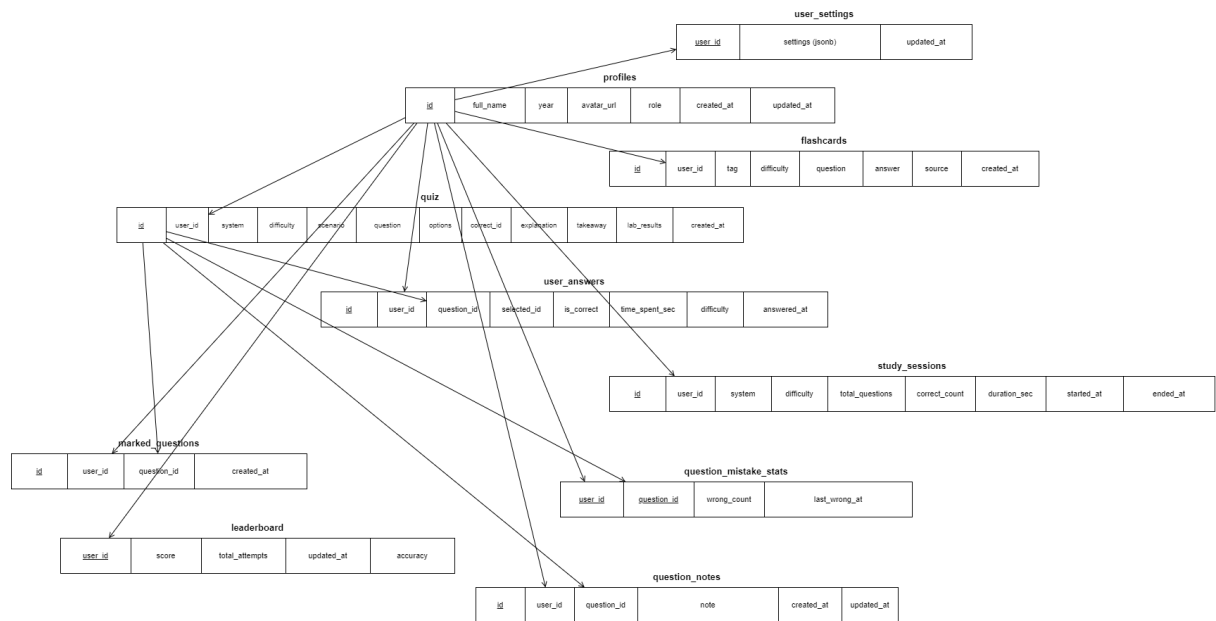
ER/EER Diagram



MedQuiz: ER/EER Diagram
Supabase public schema — conceptual view
Group - 6



Schema Diagram



Normalization

- a. Explain if your converted Schema is in 1NF or not. If not, decompose it to 1NF.
- b. Explain if your converted Schema is in 2NF or not. If not, decompose it to 2NF. Can there be any partial functional dependencies in your relational schema?
- c. Explain if your converted Schema is in 3NF or not. If not, decompose it to 3NF. Can there be any transitive dependencies in your relational schema?

Answer:

1NF: The schema is in 1NF as all attributes are atomic, single-valued, and every table has a clearly defined primary key with no repeating groups.

2NF: The schema is in 2NF as the only composite PK table (question_mistake_stats) has all non-key attributes fully dependent on the entire composite key, meaning no partial dependencies exist.

3NF: The schema is in 3NF as all non-key attributes are directly dependent on the primary key with no transitive dependencies present.

Frontend Development

Contribution of ID : 23301302, Name : Baktier Galib

Baktier Galib (23301302) contributed flashcards management for administrators, including listing system topics, editing cards, and maintaining shared decks. The screenshot should show the admin flashcards management screen at:

He contributed quiz management for administrators, covering listing, filtering, creating, and editing system questions in the Q-bank admin area. Include a view of the admin question list or an edit form at:

Finally, he contributed users management for administrators, where staff can list users, search, and adjust roles in a controlled way. The image should show the admin users table or role controls at:

Use the numbered filenames 01 through 09 under submission-screenshots/ as listed above so your report package stays consistent when you print or submit.



MANAGEMENT

Overview

Flashcard Management

Quiz Management

User Management

Logout

Super Admin
admin@medkotha.com

Quiz Management

Search question, topic, or patient story...

Refresh Filters

QUESTION	TOPIC	DIFFICULTY	STATUS	ACTIONS
What is the most appropriate initial disease-modifying therapy?	Neuroscience	Medium	System	Edit Delete
Which neurotransmitter deficiency is primarily responsible for this patient's symptoms?	Neuroscience	Medium	System	Edit Delete
Which of the following arteries is the most likely source of this hemorrhage?	Neuroscience	Hard	System	Edit Delete
Which finding most definitively confirms the diagnosis AND directly determines the next management step?	Neuroscience	Hard	System	Edit Delete
Which cranial nerve is affected?	Anatomy	Medium	System	Edit Delete
What type of enzyme inhibition does this represent?	Biochemistry	Easy	System	Edit Delete
Where does the TCA (Krebs) cycle take place within the cell?	Biochemistry	Easy	System	Edit Delete
Which enzyme is most likely deficient?	Biochemistry	Medium	System	Edit Delete
Which amino acids are affected by this transporter defect?	Biochemistry	Medium	System	Edit Delete
Which enzyme is most directly inhibited by AMPK activation, and what is the downstream metabolite whose reduction	Biochemistry	Hard	System	Edit Delete



MANAGEMENT

Overview

Flashcard Management

Quiz Management

User Management

Logout

Super Admin
admin@medkotha.com

User Management

Search accounts, view subscription status, and assign admin or student roles.

Search by email or name...

Refresh

EMAIL	NAME	ROLE	SUBSCRIPTION
admin@medkotha.com	Super Admin	Admin	None
baktier@gmail.com	Student User	Student	None
luna@gmail.com	Student User	Student	None
ckzbxmqm@burpcollaborator.net	Student User	Student	None
scrubstobe@gmail.com	Eric	Student	None
arnoldkostecki@gmail.com	Arnold	Student	None
student4@nexusmed.com	David Kim	Student	None
student1@nexusmed.com	Alice Foster	Student	None
student5@nexusmed.com	Elena Rodriguez	Student	None

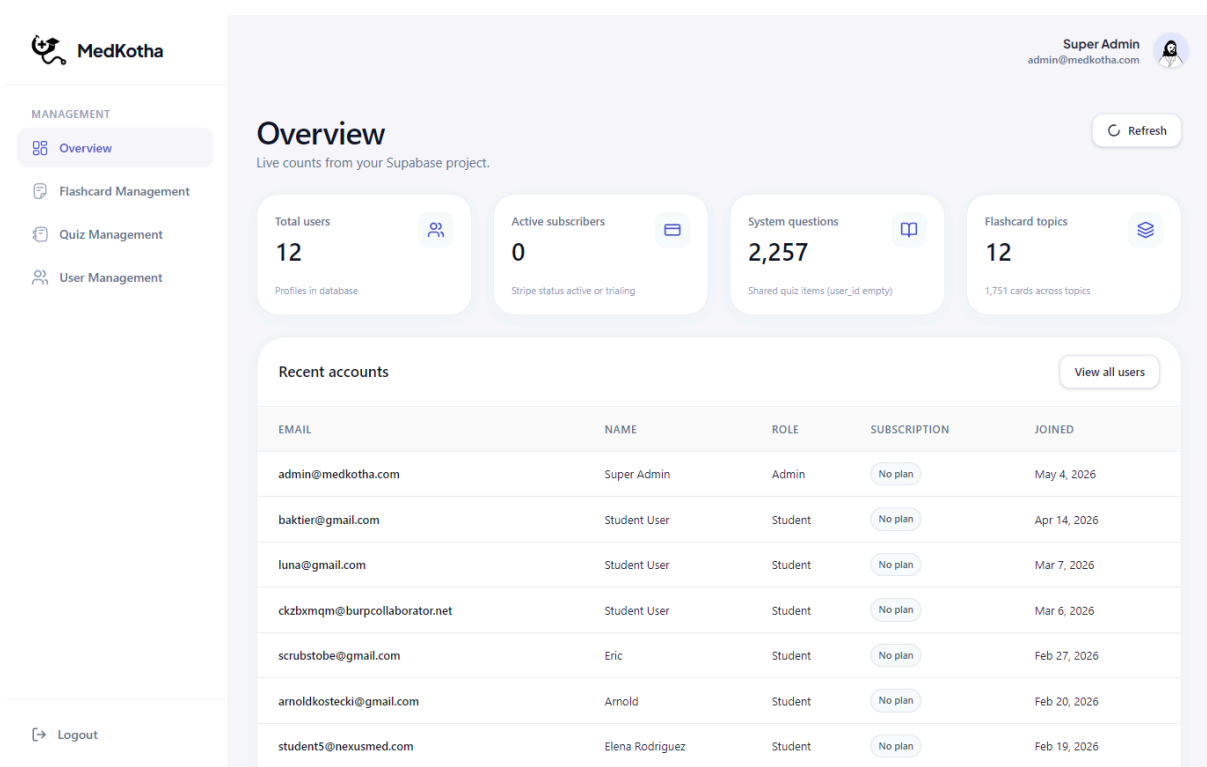
Contribution of ID : 23301489, Name : Tahsin Pathan

Tahsin Pathan (23301489) contributed the mistakes feature on the student side, which helps

users review weak areas using mistake statistics and related quiz flows. The image can show the mistakes panel, a weak spot start screen, or an explanatory empty state at:

He contributed the leaderboard page where students see ranked peers, scores, and how their own row is highlighted among others. A full-page screenshot of the leaderboard table is ideal at:

He also contributed the admin overview entry point, meaning the admin dashboard home where high-level KPIs and navigation into other admin tools appear. Show the admin overview or dashboard landing after an admin signs in at:



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STUDY

Overview

Flashcards

Quiz

ACCOUNT

Mistakes

Leaderboard

Profile

Settings

Sign Out

Weak-spot quiz

Mistake tracking

Every wrong quiz attempt is counted here. Practice only these questions to focus on weak areas.

6x WRONG

Anatomy

Which muscle is most likely injured?

Last wrong: Mar 1, 2026, 11:10 PM

Review

3x WRONG

Behavioral Science

According to Kubler-Ross, what is the typical first stage of grief?

Last wrong: Mar 2, 2026, 6:42 AM

Review

1x WRONG

Biostatistics

What is the number needed to treat?

Last wrong: Mar 7, 2026, 12:40 AM

Review

1x WRONG

Biostatistics

Which study design allows for the calculation of Relative Risk (RR)?

Last wrong: Mar 7, 2026, 12:40 AM

Review

1x WRONG

Biostatistics

What is the number of True Positives?

Last wrong: Mar 7, 2026, 12:40 AM

Review

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STUDY

Overview

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Profile

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Sign Out

Hall of Fame

Top students mastering Essential concepts. Compete to climb the ranks!

Student User

Year 1 Student

17

CORRECT

mrbeast

Resident

13

CORRECT

Eric

Year 1 Student

9

CORRECT

Arnold

Year 1 Student

0

CORRECT

David Kim

Year 1 Student

0

CORRECT

Contribution of ID : 23301334, Name : Shakeel

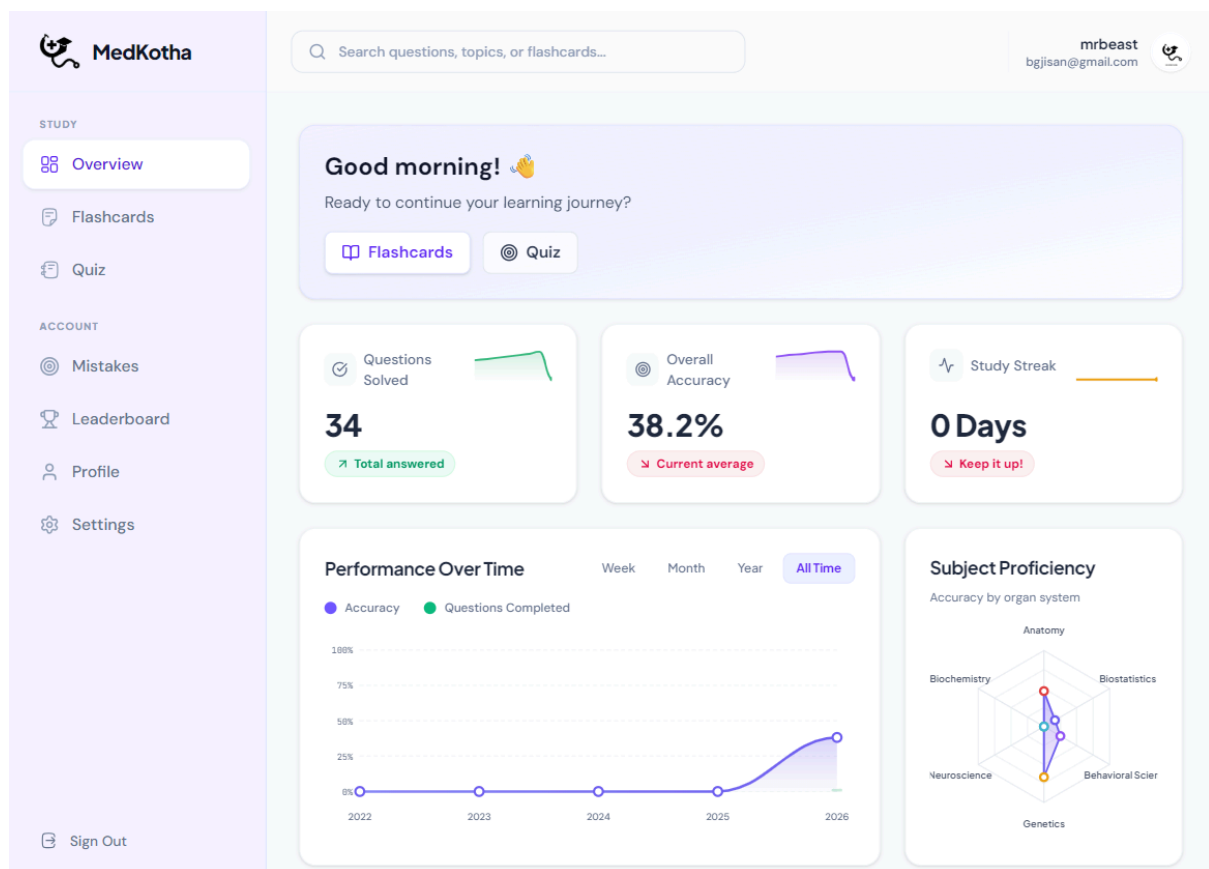
Shakeel (23301334) contributed the student overview experience on the dashboard, where

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learners see summary statistics, charts, and weekly activity at a glance. Attach or place a screenshot of the main dashboard after login at:

He also contributed the flashcards experience for students, including browsing topics and studying cards in the student flashcard flow. The screenshot should show the flashcards page or an active study session at:

His third frontend area is the quiz (Q-bank) for students, where users answer MCQs, see explanations, and move through a practice or test-style layout. Capture the screen while a question and its options are visible at:



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STUDY
 Overview
Flashcards
 Quiz

ACCOUNT
 Mistakes
 Leaderboard
 Profile
 Settings
 Sign Out

Flashcards

Select a topic (1749 Cards)

Anatomy
All Difficulties
 Easy Medium Hard

148 Cards

Behavioral Science
All Difficulties
 Easy Medium Hard

111 Cards

Biochemistry
All Difficulties
 Easy Medium Hard

150 Cards

Biostatistics
All Difficulties
 Easy Medium Hard

156 Cards

Ethics
All Difficulties
 Easy Medium Hard

112 Cards

Genetics
All Difficulties
 Easy Medium Hard

172 Cards

Immunology
All Difficulties
 Easy Medium Hard

150 Cards

Microbiology
All Difficulties
 Easy Medium Hard

150 Cards

Neuroscience
All Difficulties
 Easy Medium Hard

150 Cards

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STUDY
 Overview
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Quiz

Free Preview (2255 Questions)

Anatomy
Organ System
 3 / 294 completed

294 Qs

Behavioral Science
Organ System
 1 / 186 completed

186 Qs

Biochemistry
Organ System
 1 / 150 completed

150 Qs

Biostatistics
Organ System
 6 / 186 completed

186 Qs

Ethics
Organ System
 0 / 206 completed

206 Qs

Genetics
Organ System
 3 / 182 completed

182 Qs

Immunology
Organ System
 1 / 173 completed

173 Qs

Microbiology
Organ System
 1 / 186 completed

186 Qs

Neuroscience
Organ System
 2 / 192 completed

192 Qs

Create Custom Session

Backend Development

Contribution of ID : 23301302, Name : Baktier Galib

Baktier Galib (23301302) owns the flashcards management backend for shared content, which depends on system-owned rows (user_id null where the schema allows), admin Row Level Security policies and helpers in supabase/sql/admin_system_content.sql, and the admin_system_flashcard_topics RPC, while api.ts provides typed helpers such as insertAdminSystemFlashcard and updateAdminSystemFlashcard for the admin UI.

For quiz management, the same stack reuses the quiz table with admin-specific policies from the admin SQL bundle, and the admin CRUD paths in api.ts (insertAdminSystemQuestion, updateAdminSystemQuestion, deleteAdminSystemQuestion, and listing helpers) perform the inserts, updates, and deletes that PostgREST translates into SQL for system questions only.

For users management, the admin_list_users and admin_set_user_role functions in supabase/sql/admin_user_management.sql join profiles to auth.users and update the role column safely for admins, and the front end calls them through fetchAdminUsers and adminSetUserRole in api.ts.

Contribution of ID : 23301489, Name : Tahsin Pathan

Tahsin Pathan (23301489) implemented the mistakes backend around the question_mistake_stats table and the trigger defined in CSE370/project/supabase/sql/09_question_mistake_stats.sql, which updates counts when rows are inserted into user_answers. The application reads those aggregates through fetchMistakeStats in api.ts, so wrong answers are reflected without manual client-side tallying.

For the leaderboard, the cached leaderboard table, refresh logic, triggers, and the get_leaderboard RPC are defined in 10_leaderboard.sql, and the UI loads results through fetchLeaderboard in api.ts and LeaderboardPage.tsx, which again resolves to SQL execution on the server.

For admin overview, KPI-style aggregates are returned by the admin_dashboard_stats function in supabase/sql/admin_dashboard_stats.sql, which checks admin privileges before returning JSON, and the SPA invokes it via fetchAdminDashboardStats in api.ts.

Contribution of ID : 23301334, Name : Shakeel

Shakeel (23301334) implemented and integrated the student overview backend by relying on Supabase RPCs such as `get_dashboard_stats`, `get_weekly_activity`, `get_subject_performance`, and `get_activity_history`, whose definitions live in SQL files under the workspace (for example `supabase/sql/dashboard_stats.sql` and related patches). The React layer calls these functions through `fetchDashboardStats` and related helpers in `CSE370/project/src/lib/api.ts`, and `DashboardPage.tsx` orchestrates loading and display.

For flashcards, his work connects to the `flashcards` table and its Row Level Security in `CSE370/project/supabase/sql/02_flashcards.sql`, while topic aggregation for students uses the `get_flashcard_topics` function in `11_helpers_and_topics.sql`. The TypeScript side exposes this through functions such as `fetchFlashcards` and `fetchFlashcardTopics` in `api.ts`.

For the quiz, persistence uses the `quiz` and `user_answers` tables with policies in `03_quiz.sql` and `04_user_answers.sql`. Submitting attempts goes through `submitAnswer` in `api.ts`, loading banks uses `fetchQuestionsBySystem` and related queries, and topic summaries use `get_question_topics` or the fallback logic in the same module, all of which ultimately execute as SQL on Postgres when students use `QbankPage.tsx`.

Source Code Repository

[MedKotha Source Code](#)

Conclusion

MedKotha shows that a full-featured study product can be delivered with a modern React front end and a managed PostgreSQL backend through Supabase, without maintaining a custom application server for every CRUD path. Database rules, triggers, and stored procedures keep aggregation and sensitive admin operations on the server, while the SPA focuses on usability, navigation, and charts. The work was split so that student-facing study flows, analytics and competition features, and administrative content and user management each have clear owners on both the UI and the data layer. Together, the architecture supports secure multi-user access, evolving question banks, and a path to extend billing or notifications as the product grows. For course purposes, the main takeaway is that backend logic is visible in SQL migrations and RPC definitions, and the submission screenshots plus this text document map each teammate's features to those technical artifacts.

References

[1] Supabase documentation - overview, Auth, Database, Row Level Security, and JavaScript client. <https://supabase.com/docs>

[2] PostgreSQL documentation - SQL language, triggers, and stored procedures. <https://www.postgresql.org/docs/>

[3] React documentation - UI library used for the student and admin apps. <https://react.dev/>